

FRONTONE, Italy

WMO: 161790

Lat: 43.517N

Lon: 12.728E

Elev: 574

StdP: 94.62

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.0	-12.0	1.4	2.6	-9.8	1.7	2.7	17.2	9.1	14.6	9.2	3.0	250	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 34.0	(d) 19.6	(e) 32.2	(f) 19.4	(g) 30.6	(h) 19.0	(i) 21.9	(j) 29.0	(k) 21.0	(l) 28.2	(m) 20.2	(n) 27.5	(o) 3.7	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	15.7	25.1	18.7	14.5	24.2	17.7	13.6	23.5	66.9	28.7	63.5	28.0	60.8	27.4	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	9.9	8.0	-5.0	36.3	1.9	2.0	-6.4	37.7	-7.5	38.8	-8.6	39.9	-10.0	41.3
			-6.3	23.8	1.6	0.9	-7.5	24.5	-8.4	25.1	-9.3	25.6	-10.5	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	4.7	5.0	8.4	12.9	16.8	21.9	24.8	24.3	18.8	14.3	9.5	5.9	(6)
(7)		DBStd	8.01	3.67	3.99	4.21	3.30	3.93	4.21	3.38	3.55	3.75	3.62	3.61	3.53	(7)
(8)		HDD10.0	593	167	143	82	10	1	0	0	0	0	8	51	132	(8)
(9)		HDD18.3	2185	422	373	309	166	79	15	2	2	37	132	264	384	(9)
(10)		CDD10.0	2052	3	3	31	96	211	357	459	444	266	140	37	6	(10)
(11)		CDD18.3	602	0	0	0	2	31	122	202	187	52	6	0	0	(11)
(12)		CDH23.3	4725	0	0	0	4	156	961	1811	1553	233	7	0	0	(12)
(13)		CDH26.7	1737	0	0	0	0	28	325	719	612	53	0	0	0	(13)
(14)	Wind	WSAvg	2.8	3.1	3.2	3.2	2.7	2.6	2.5	2.7	2.4	2.5	2.3	3.0	3.3	(14)
(15)	Precipitation	PrecAvg	1198	82	94	93	106	101	86	67	89	104	116	154	110	(15)
(16)		PrecMax	1874	175	206	178	249	273	174	214	323	218	270	342	196	(16)
(17)		PrecMin	652	8	21	16	24	2	17	12	2	20	1	56	21	(17)
(18)		PrecStd	278	50	60	49	57	62	46	47	73	68	76	64	49	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	15.1	19.7	23.9	29.2	34.1	36.0	36.6	30.9	24.1	18.1	14.7	(19)
(20)			MCWB	7.8	7.9	10.7	13.2	17.7	20.1	19.4	19.4	17.9	15.6	14.3	11.3	(20)
(21)		2%	DB	12.1	13.0	17.5	21.7	27.0	32.1	34.0	34.1	28.1	22.1	16.6	13.2	(21)
(22)			MCWB	8.3	7.7	10.2	13.2	17.2	19.9	19.6	19.4	18.2	16.0	13.2	10.2	(22)
(23)		5%	DB	11.1	11.4	15.6	19.9	24.9	30.1	32.3	31.9	26.0	20.2	15.3	12.0	(23)
(24)			MCWB	7.9	7.4	9.7	12.1	16.4	19.1	19.5	19.0	17.3	15.3	12.2	9.3	(24)
(25)		10%	DB	9.9	10.1	14.0	18.1	23.1	28.3	30.8	30.1	24.1	18.9	14.1	10.8	(25)
(26)			MCWB	7.2	6.9	8.9	11.4	15.2	18.4	19.1	18.6	16.8	15.0	11.3	8.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.6	12.7	16.4	20.3	23.0	23.6	22.5	20.9	18.4	16.2	12.1	(27)
(28)			MCDWB	12.0	12.7	16.5	20.3	26.2	29.6	30.0	29.5	26.2	21.0	17.0	13.8	(28)
(29)		2%	WB	9.1	9.3	11.4	14.5	18.4	21.4	21.9	21.5	19.5	17.3	14.1	10.7	(29)
(30)			MCDWB	11.4	11.3	15.1	18.8	24.6	28.5	29.3	28.4	24.5	20.1	15.6	12.5	(30)
(31)		5%	WB	8.4	8.4	10.5	13.3	17.1	20.3	21.1	20.6	18.7	16.3	12.8	9.6	(31)
(32)			MCDWB	10.5	10.2	14.0	17.5	23.0	27.8	28.8	27.7	23.3	18.8	14.4	11.4	(32)
(33)		10%	WB	7.4	7.4	9.6	12.4	15.8	19.2	20.2	19.9	17.9	15.5	11.7	8.6	(33)
(34)			MCDWB	9.4	9.4	13.0	16.4	21.8	26.8	27.8	27.2	22.4	17.8	13.6	10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.6	6.0	7.5	8.3	8.9	9.4	9.1	7.3	5.5	4.3	4.0	(35)
(36)			MCDBR	5.0	5.9	7.9	9.2	9.8	10.2	10.6	10.5	9.3	6.9	5.3	4.6	(36)
(37)			MCWBR	3.7	4.1	4.6	5.5	6.0	5.7	5.3	5.0	4.6	4.0	3.8	3.7	(37)
(38)		5% WB	MCDBR	4.7	5.0	6.9	8.1	9.3	9.5	9.5	9.2	8.0	6.0	4.7	4.2	(38)
(39)			MCWBR	3.8	4.1	4.5	5.3	6.1	5.8	5.4	5.2	4.8	4.0	4.0	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.339	0.368	0.401	0.403	0.403	0.395	0.392	0.381	0.366	0.342	0.318	(40)	
(41)		taud	2.460	2.386	2.318	2.236	2.272	2.308	2.330	2.355	2.375	2.411	2.421	2.483	(41)	
(42)		Ebn at Noon	809	851	867	863	870	868	871	862	844	808	769	777	(42)	
(43)		Edh at Noon	74	93	113	132	131	127	123	116	106	90	76	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.43	2.23	3.42	4.58	5.69	6.58	6.82	5.89	4.19	2.70	1.61	1.28	(44)	
(45)		RadStd	0.20	0.30	0.48	0.45	0.53	0.35	0.34	0.41	0.28	0.28	0.18	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	-77	+83	N/A

Nomenclature: See separate page